

- (b) Figure 6.2 shows the profile of the water level along line L at a certain instant. Sketch on the same figure the profile at a time $\frac{T}{2}$ later, where T is the period of the water waves. (1 mark)

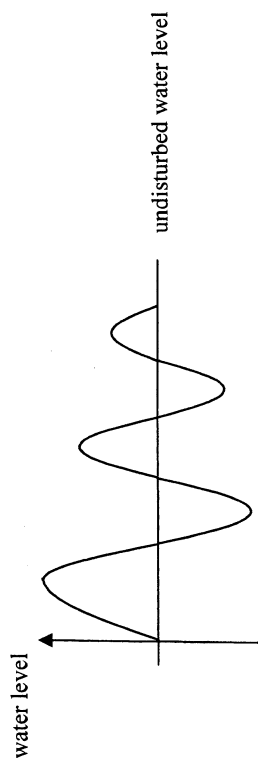


Figure 6.2

- (c) Q is a point on the line joining S_1 and S_2 as shown in Figure 6.1. State the kind of interference that occurs at Q and give a reason for this occurrence. (2 marks)

Answers written in the margins will not be marked.

- *(d) A similar double-slit set-up is used for the demonstration of the interference of light in which the separation between slits S_1 and S_2 is 0.5 mm and the screen is at 2.5 m from the slits. Calculate the average separation between adjacent bright fringes on the screen for a monochromatic light of wavelength 550 nm. (2 marks)

Answers written in the margins will not be marked.

7. A luminous object PQ is placed 15 cm in front of a convex lens AB as shown in Figure 7.1.

(a) The focal length of the lens is 5 cm.

(i) Use a graphical method to find the location of the image of the object. Clearly draw all the construction lines on Figure 7.1 and state the nature of the image. (4 marks)

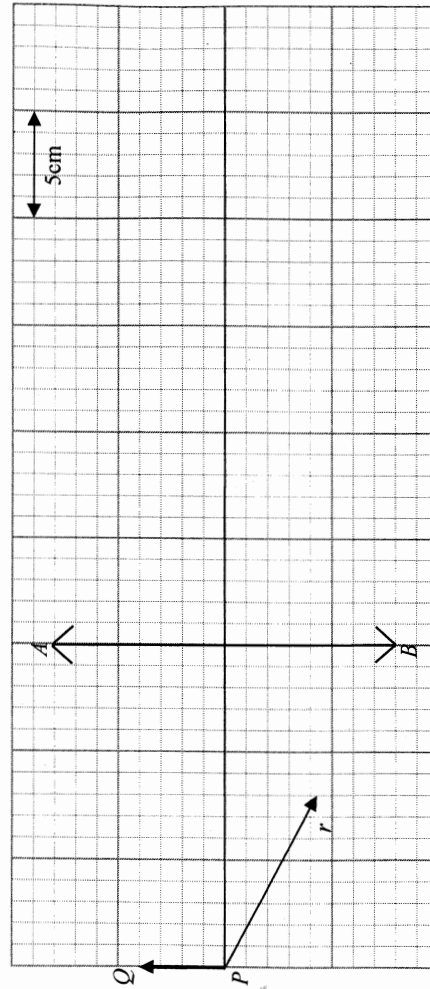


Figure 7.1

(ii) Complete the path of ray r on Figure 7.1 to show how it travels after passing through the convex lens. (1 mark)

(b) Suppose that a convex lens of focal length 10 cm is used instead while the size of the lens and the object distance of PQ from the lens remain unchanged.

*(i) Use the lens formula to find the image distance. Find also the linear magnification of the image. (3 marks)

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(2 marks)

(ii) Compare the brightness of this image with that in (a). Explain.

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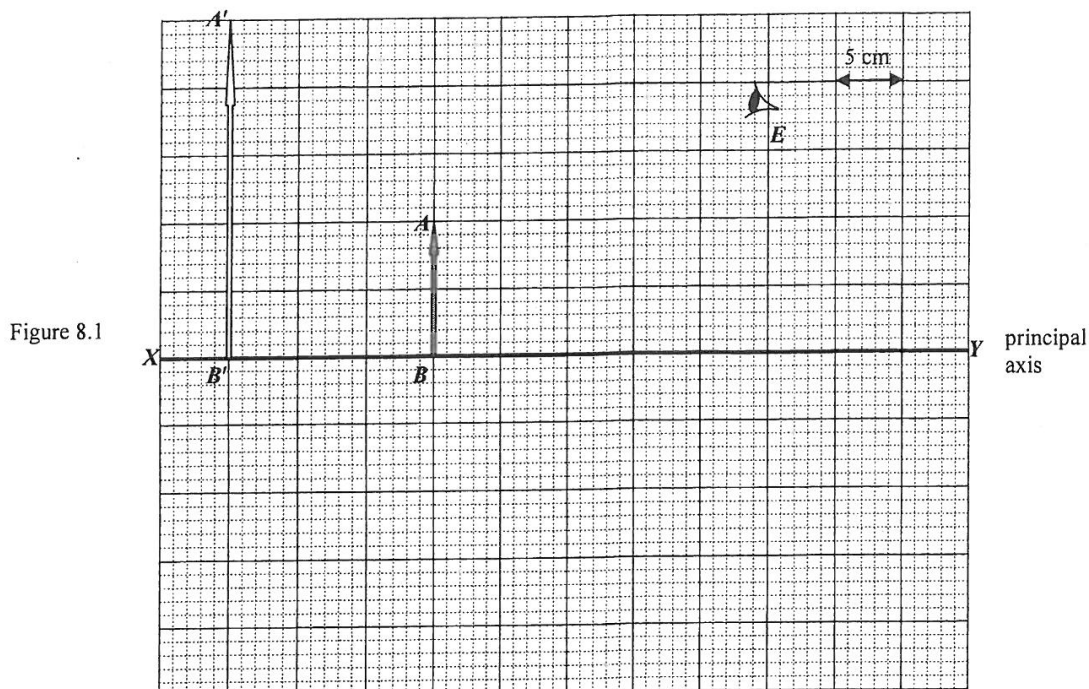
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8. In Figure 8.1, $A'B'$ represents the image of an object AB formed by a lens L (not shown) where XY is the principal axis of the lens.



- (a) (i) Is the image real or virtual ? (1 mark)

- (ii) What kind of lens is used ? Explain your answer. (2 marks)

- (b) (i) Locate the optical centre O of lens L and draw on Figure 8.1 the position of lens L . (1 mark)

- (ii) By drawing an additional light ray, mark the principal focus F of the lens and find its focal length. The horizontal scale is 1 cm to 5 cm. (2 marks)

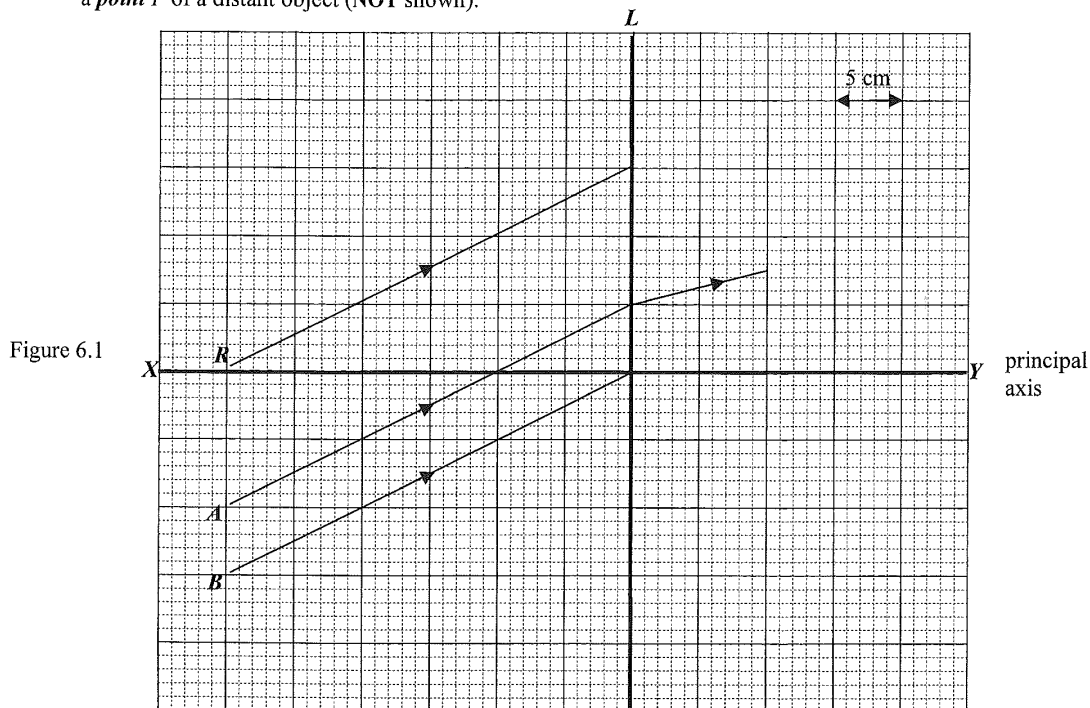
Focal length =

- (c) Draw a light ray to show how the eye E shown can see the image of head A through lens L . (2 marks)

- (d) State an application of lens L in the situation as shown above. (1 mark)

Answers written in the margins will not be marked.

6. In Figure 6.1, XY is the principal axis of a thin spherical lens L while A, B are two parallel rays coming from a point P of a distant object (NOT shown).



- (a) What kind of lens is L ? Explain. (2 marks)

- (b) (i) Locate the image of P (denoted it as point P'). (2 marks)

- (ii) Hence, determine the focal length of the lens. (1 mark)

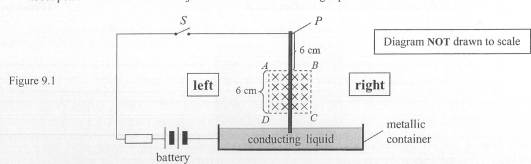
Focal length =

- (c) R is a ray coming from the same point P ; complete its path after passing through the lens. (1 mark)

- (d) Based on the situation shown in the ray diagram above, describe a simple experimental method to determine the focal length of lens L . (2 marks)

Answers written in the margins will not be marked.

9. Figure 9.1 shows a set-up for demonstrating one of Faraday's discoveries. A light metal rod is free to rotate about point P while its lower end just touches some conducting liquid in a metallic container.



A uniform magnetic field pointing into the paper is applied over the region $ABCD$ containing part of the rod. When switch S is closed, the rod 'kicks' out and leaves the liquid surface.

(a) State the direction (to the left / to the right / into the paper / out of the paper) that the rod 'kicks' and describe the subsequent motion of the rod. (3 marks)

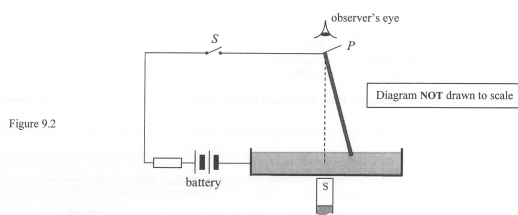
(b) When switch S is closed, the initial moment about point P that makes the rod 'kick' out is $7.2 \times 10^{-4} \text{ N m}$. Assume that the magnetic force acts at the midpoint of the part of the rod within the magnetic field.

(i) Calculate the magnetic force acting on the rod at that instant. (2 marks)

(ii) Hence, find the strength B of the magnetic field if the current flowing through the rod is 3.2 A when the circuit is closed. (2 marks)

Answers written in the margins will not be marked.

(c) Now the uniform magnetic field is removed and a bar magnet is placed underneath the container as shown in Figure 9.2. The rod is held tilted at an angle to the vertical but with its lower end still in the conducting liquid.

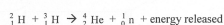


(i) Sketch on Figure 9.2 the field lines around the rod due to the bar magnet. (1 mark)

(ii) After closing switch S and the rod is released from rest, describe its subsequent motion viewed from above. (1 mark)

Answers written in the margins will not be marked.

10. Scientists had been experimenting controlled fusion in a nuclear reactor in which deuterium (${}^2_1\text{H}$) and tritium (${}^3_1\text{H}$) undergo the following nuclear fusion:



Given:

mass of a deuterium nucleus	= 2.014102 u
mass of a tritium nucleus	= 3.016049 u
mass of a helium nucleus	= 4.002602 u
mass of a neutron	= 1.008665 u

- *(a) Calculate the energy released, in MeV, in the above nuclear fusion. (2 marks)

- (b) A deuterium nucleus and a tritium nucleus have to be within 10^{-15} m for nuclear fusion to occur and that a large amount of work done (about 0.4 MeV) is needed to bring two well separated nuclei to such a close distance.

- (i) Explain why a large amount of work done is needed and state the kind of energy this work done has become. (2 marks)

In the nuclear reactor, deuterium and tritium exist as plasma, which is a mixture of ions at a very high temperature.

- (ii) Explain why a very high temperature is needed for nuclear fusion to occur. (1 mark)

- *(iii) Estimate the order of magnitude of the minimum temperature at which fusion of deuterium and tritium nuclei would be possible if the plasma can be regarded as an ideal gas. (Hint: For an ideal gas, the gas molecules each is assumed to have an average kinetic energy $E_K = \frac{3kT}{2N_A}$) (2 marks)

END OF PAPER

Sources of materials used in this paper will be acknowledged in the *Examination Report and Question Papers* published by the Hong Kong Examinations and Assessment Authority at a later stage.

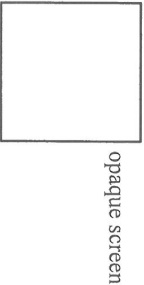
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(ii) Sketch the image on the screen seen by the observer.

(1 mark)



(b) The separation D is varied while the position of the lens is adjusted to form an image once again on the screen. The corresponding object distance u is obtained for plotting a graph of D against u (Figure 5.2).

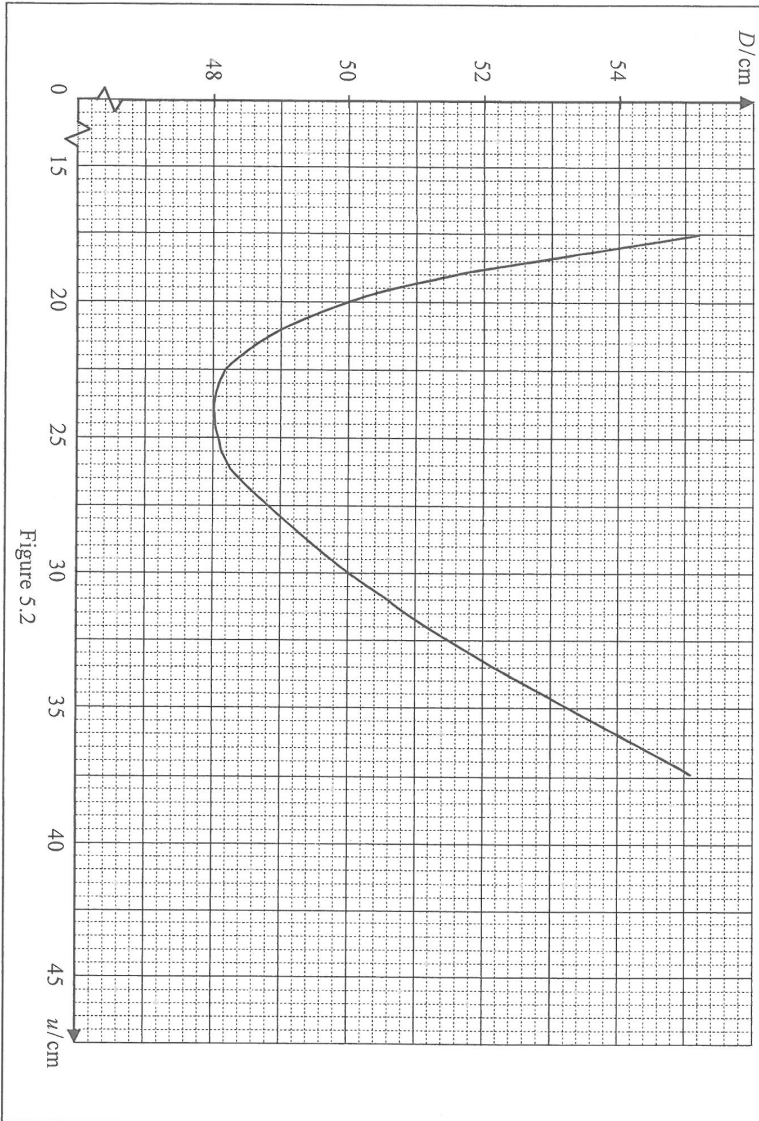
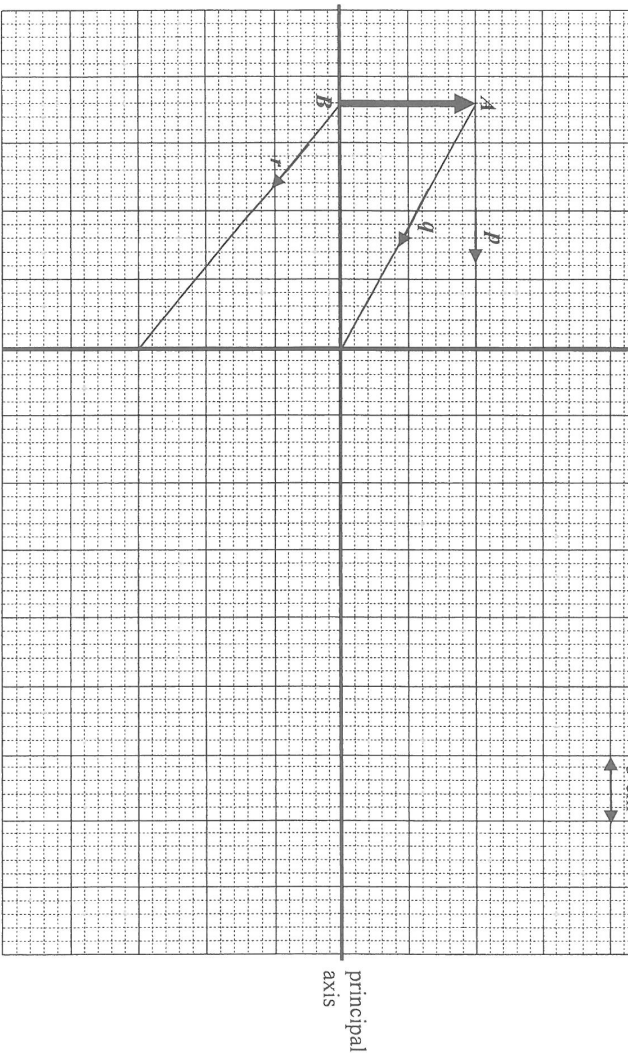


Figure 5.2

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- (ii) Indicate the image formed by AB (denote it as I) and draw the refracted rays of p , q and r . (3 marks)
- (iii) Hence find the focal length of the lens. The horizontal scale is 1 cm to 5 cm. (1 mark)

Focal length =

- (iv) Keeping the object and screen fixed in position, suggest where Kitty should move the lens to such that an image can be formed again on the screen. State the ratio $\frac{\text{height of this new image}}{\text{height of the original image}}$. (2 marks)

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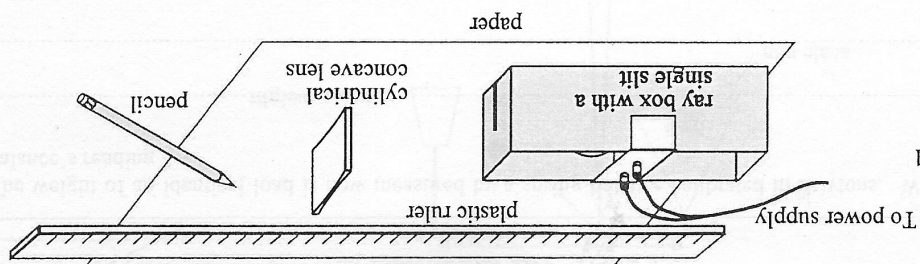
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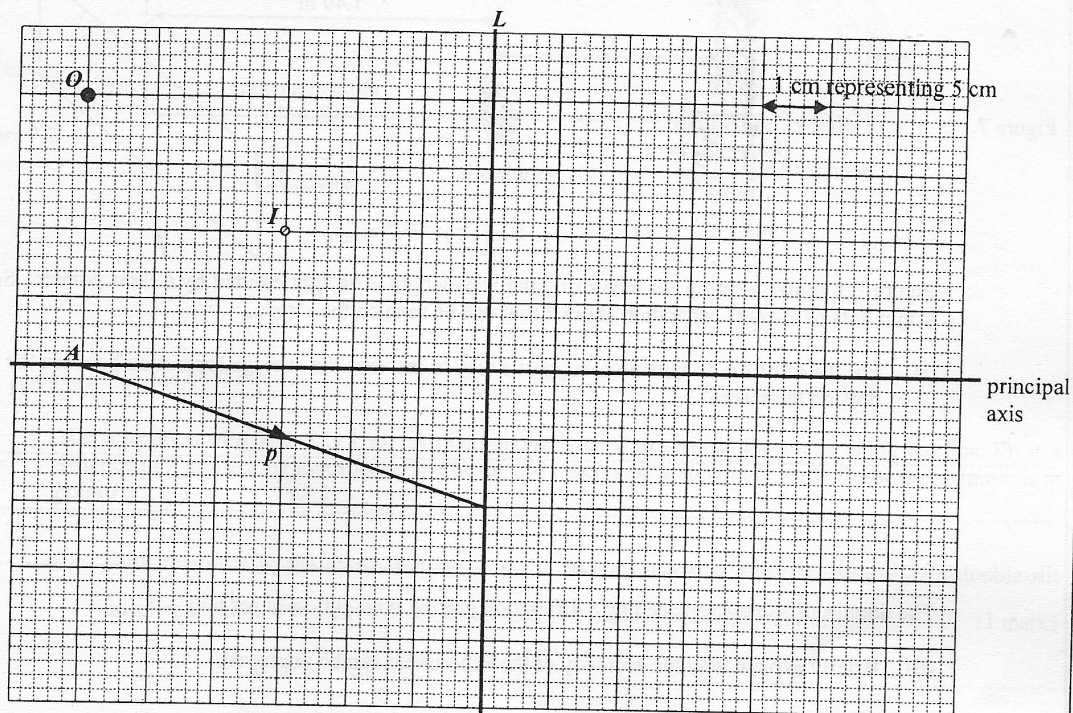
Describe how you would use the above apparatus to find the focal length of the lens and state ONE possible source of error in the experiment. (5 marks)



6. (a) You are given a ray box with a single slit (producing a fine light beam), a cylindrical concave lens, a plastic ruler, a pencil and a piece of paper as shown in Figure 6.1.

Answers written in the margins will not be marked.

- (b) In the figure below, L represents another cylindrical lens. A vertical pin used as the object is placed at O , the image is formed at I by the lens. The horizontal scale is 1 cm to 5 cm.



- (i) What kind of lens is used ? Explain.

(2 marks)

- (ii) Draw a suitable light ray to locate the principal focus F of lens L . Find its focal length.

Focal length =

(2 marks)

- (iii) Complete the path for the ray p from point A .

(1 mark)

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6. In Figure 6.1, AB represents the virtual image of an object formed by lens L . The magnification of the image is 0.4. The horizontal scale is 1 cm to 5 cm.

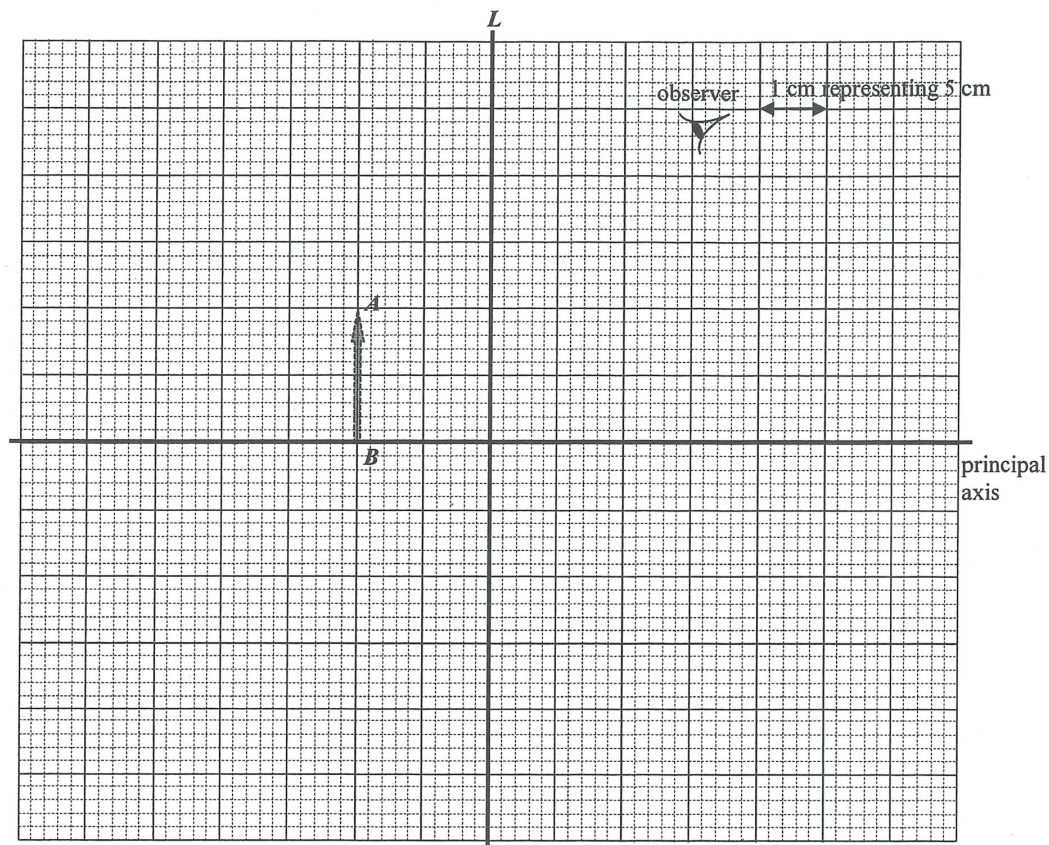


Figure 6.1

- (a) What kind of lens is used? Explain. (2 marks)

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- (b) Indicate on Figure 6.1 the position and height of the object. (2 marks)

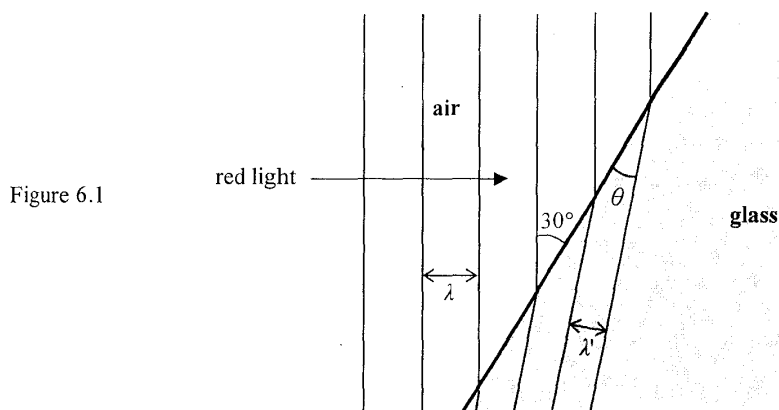
- (c) By drawing a suitable light ray, locate and mark the position of the focus, F , of the lens. Find the focal length of the lens. (3 marks)

Focal length =

- (d) Draw a light ray emerging from the object to illustrate how the observer in the figure can see the tip A of the image. (2 marks)

Answers written in the margins will not be marked.

6. (a) When red light of wavelength $\lambda = 675 \text{ nm}$ is incident at an angle of 30° from air to glass as shown in Figure 6.1, refraction occurs such that its wavelength in glass becomes $\lambda' = 450 \text{ nm}$ while the angle of refraction is θ .



- (i) What is the frequency of the red light in glass ? (2 marks)

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- (ii) Find θ . (2 marks)

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- (iii) If the red light is replaced by blue light, θ will decrease. Compare the refractive index of glass for the red light and blue light. (1 mark)

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6. (b) Figure 6.2 shows a system for projecting a transparent photographic slide O onto a screen. The slide and the screen are 1 m apart. A beam of white light illuminates the slide. The position of lens L is adjusted until a sharp image of O magnified linearly to 9 times is formed on the screen.

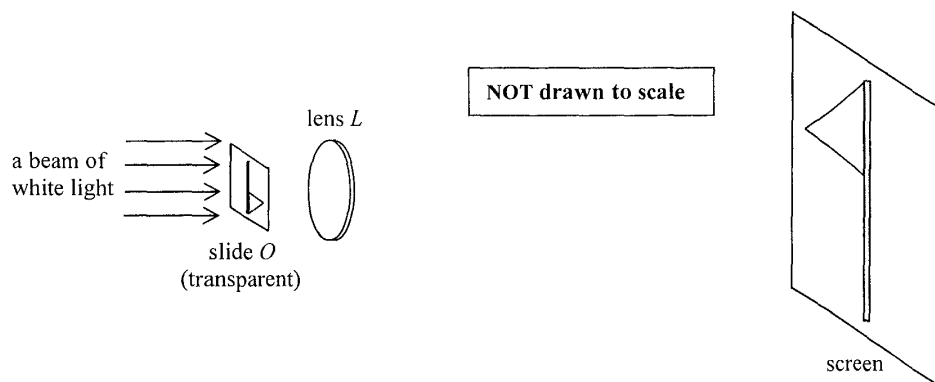
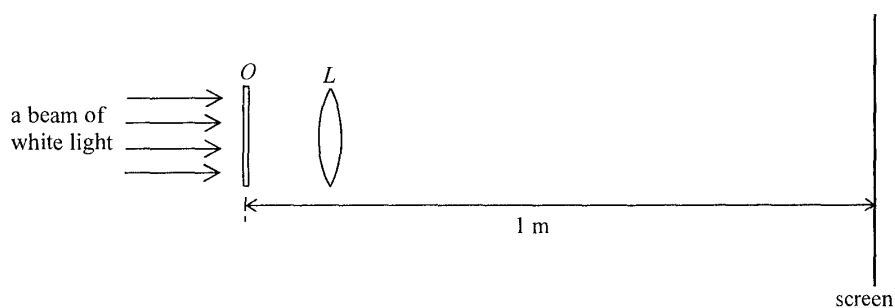


Figure 6.2



- (i) State the nature of this magnified image. (1 mark)

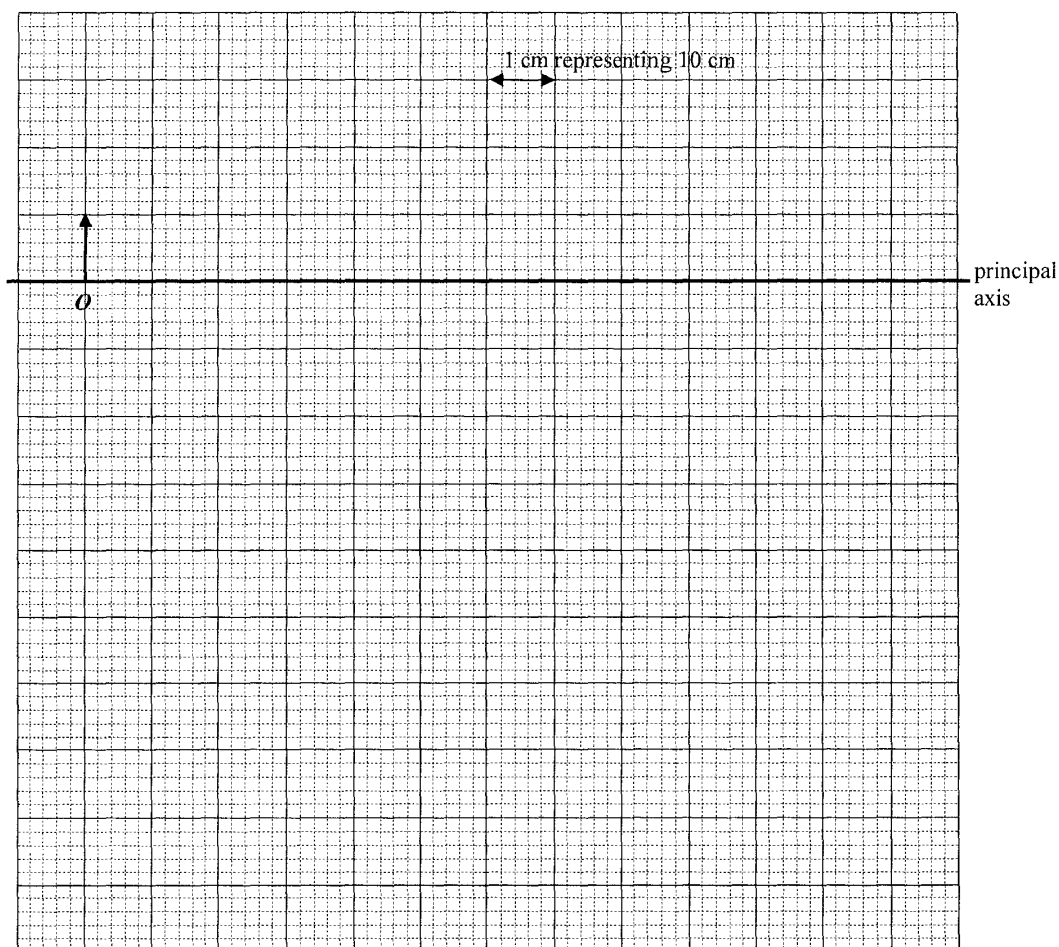
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- (ii) Find the separation between O and L . (1 mark)

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- (iii) Draw a ray diagram to show how the image of slide O is formed on the screen. Locate the focus F of lens L on your diagram and find the focal length of L . (Slide O and the principal axis of the lens have been drawn for you.) (5 marks)

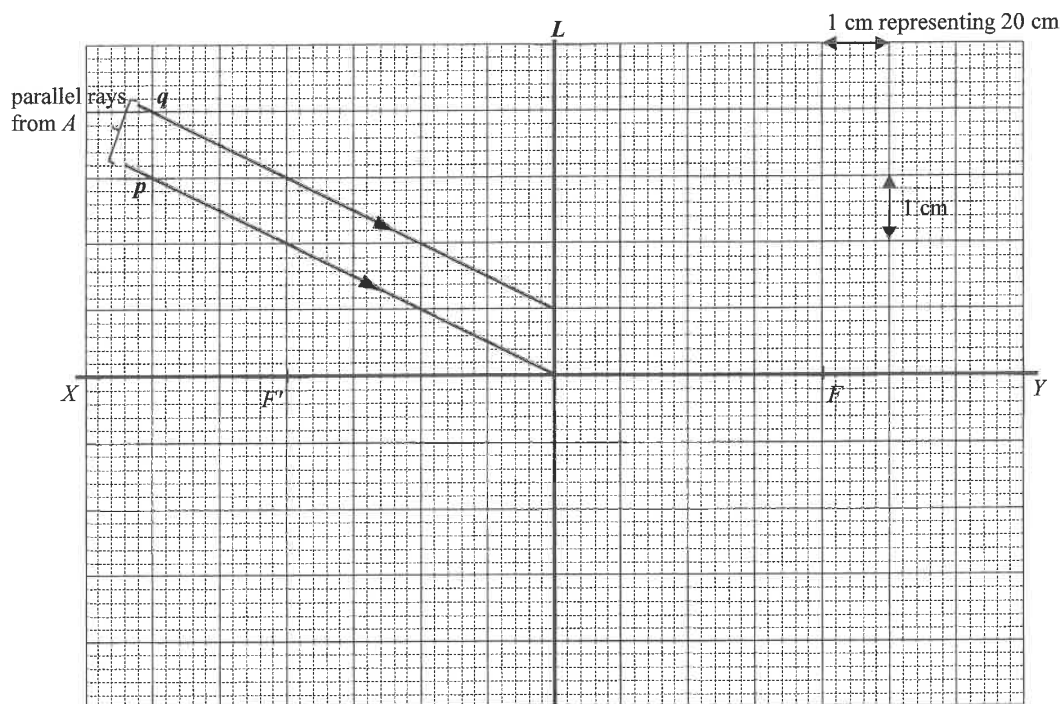


Focal length of L =

- (iv) When a black-and-white slide is projected onto the screen, the image has colour edges. Explain briefly. (Hint: the lens is made of glass.) (2 marks)

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5. In the figure below, XY is the horizontal principal axis of a convex lens L with principal foci F and F' . p, q are parallel rays coming from point A of a distant object AB . (The object can be represented by a vertical arrow $\begin{smallmatrix} A \\ \uparrow \\ B \end{smallmatrix}$ but it is **not** shown on the figure and its end B is on the principal axis).



- (a) (i) Draw the refracted rays of p and q so as to locate the image of A (denoted as A'). Hence mark the image $A'B'$ of object AB . (3 marks)
- (ii) Suggest an experiment to verify whether a real image is formed in the above situation. (2 marks)

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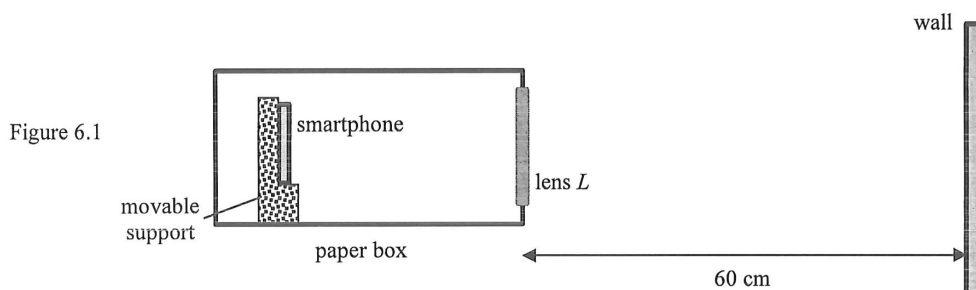
- (b) (i) Use the ray diagram drawn to estimate the ratio $\frac{\text{height of object } AB}{\text{distance of } AB \text{ from } L}$. The horizontal and vertical scales are 1:20 and 1:1 respectively. (2 marks)

- (ii) Hence, estimate the height of object AB which is a lamp post at a distance of 200 m from lens L . (1 mark)

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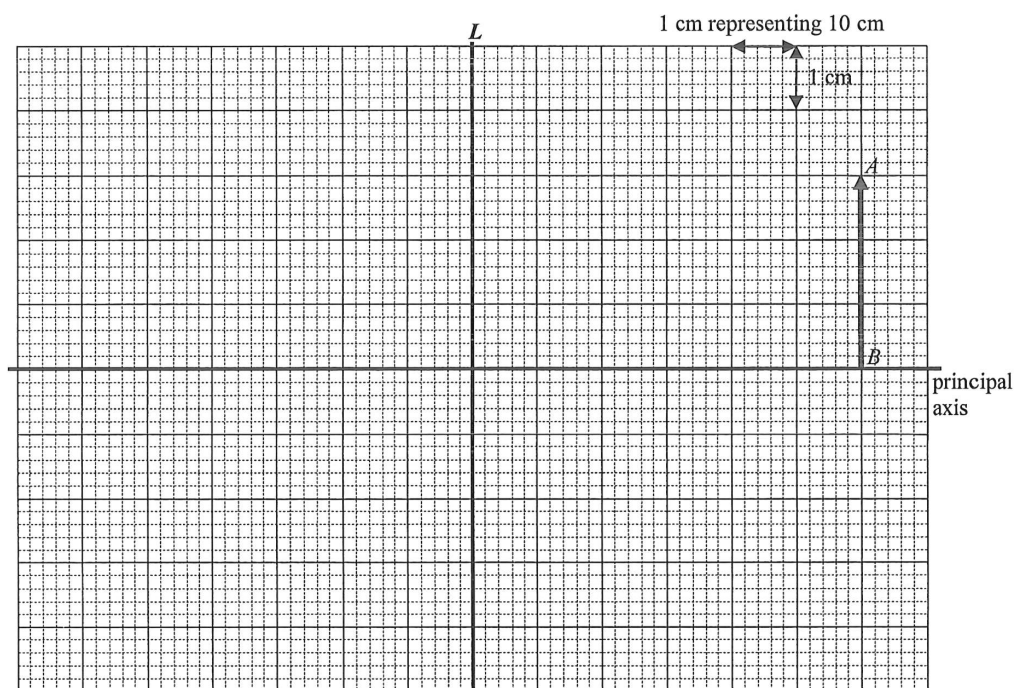
6. A student builds a smartphone projector as shown in Figure 6.1. It consists of a paper box with a lens L in front and a movable smartphone support inside. The position of the smartphone can be adjusted until a sharp image is formed on a wall.



- (a) What kind of lens is used? Explain briefly.

(2 marks)

- (b) In the figure below, AB represents the image formed on the wall, which is 60 cm from lens L . Given that the magnification of the image is 3.



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- (i) Locate and draw the object (i.e. the display on the smartphone) using an arrow and denote it as O . (1 mark)
- (ii) Complete the ray diagram by drawing **TWO** light rays reaching A from the object. (2 marks)
- (iii) Hence, or otherwise, find the focal length of L . (1 mark)

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- (c) If the separation between lens L and the wall is slightly increased, state the adjustment to be made so as to restore a sharp image on the wall. (1 mark)

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- (d) Suggest a way to minimise unwanted reflections inside the box. (1 mark)

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